Transformers as Measure Learners: Wasserstein Scaling Laws for KV Cache Compression

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Abstract

In long-context In-Context Learning (ICL), the main bottleneck is the Key–Value (KV) cache, whose memory grows linearly with sequence length. Existing compression methods prune tokens or quantize values, but treat the context as a scored sequence and offer little guidance on what *optimal* compression should be.

We instead view the context as an empirical probability measure and the KV cache as its finite-support approximation. Building on universality results for Transformers on measures, we show that ICL predictions are Lipschitz with respect to the Wasserstein-1 distance between context measures, which yields a geometric error bound for KV compression. Combining this with quantization theory gives a minimax-optimal $m^{-1/d}$ scaling of ICL error in the KV budget m, where d is the intrinsic feature dimension.

Guided by this view, we propose **Measure-Aware Compression (MAC)**, which approximates Wasserstein minimization via weighted K-Means at inference time and Sinkhorn-regularized induced attention during training. On synthetic ICL tasks and the LUNA16 medical benchmark, MAC achieves better memory–accuracy tradeoffs than strong KV baselines such as H2O and Quest.

1. Introduction

Large Transformer models have emerged as powerful *in-context learners*: given a long context of input–output examples, they can synthesize a predictor for new queries without updating parameters (Brown et al., 2020; Garg et al., 2022). With recent progress in long-context architectures such as LongRoPE, Unlimiformer, and proprietary systems like Gemini 1.5 and Llama 2 (Ding et al., 2024; Bertsch et al., 2024; Team et al., 2024; Touvron et al., 2023), context windows have reached millions of tokens, enabling foundation models for domains like medicine to reason over entire patient trajectories, imaging studies, or clinical records (Moor et al., 2023).

The KV Memory Wall of Long-Context ICL. Despite architectural advances, long-context In-Context Learning (ICL) on Transformers is fundamentally constrained by the *Key–Value (KV) cache*. In standard self-attention (Vaswani et al., 2017), the KV cache grows linearly $O(N)$ with sequence length N, and attention computation scales as $O(N^2)$. For million-token contexts, storing KV states in FP16 can require hundreds of gigabytes of memory, exceeding typical GPU budgets and making inference prohibitively slow—especially in resource-constrained environments such as hospital edge servers or autonomous labs. Alternative sequence models, including linear/streaming attention (Katharopoulos et al., 2020) and state-space models like Mamba (Gu & Dao, 2024), reduce asymptotic complexity, but empirical evidence suggests that standard Transformers remain highly competitive ICL systems due to their flexible associative memory (Bietti et al., 2024). This motivates a principled approach to *compressing* the KV cache while preserving ICL performance.

Heuristic KV Compression and Its Limits. A rapidly growing literature attacks the KV bottleneck with engineering techniques. Token-pruning methods such as H2O, Scissorhands, SnapKV, and PyramidKV (Zhang et al., 2024b; Liu et al., 2024a; Li et al., 2024; Zhang et al., 2024a; Ge et al., 2024) retain "important" tokens based on accumulated attention scores or layer-wise saliency and evict the rest. Retrieval-based approaches, including Quest, Infini-attention, RetrievalAttention, and Unlimiformer (Tang et al., 2024; Munkhdalai et al., 2024; Gao et al., 2024; Bertsch et al., 2024), maintain a larger external memory and fetch relevant blocks on demand. Quantization and distillation-based methods like KIVI, attention sinks, and KV distillation (Liu et al., 2024b; Xiao et al., 2024; Nawrot et al., 2024) instead compress the representation of each KV pair to a few bits or a small distilled cache.

All of these methods, however, share two structural limitations. First, they treat the context as an *ordered sequence with local scores* (per-token or per-block importance) rather than as a global geometric object. Second,

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they lack *distribution-level guarantees*: there is no theory explaining what an *optimal* compressed KV cache should look like, or how the induced compression error scales with the memory budget and the structure of the underlying data.

Our Perspective: Transformers as Measure Learners. We propose a different view. Instead of thinking of the context as a scored sequence, we interpret it as an *empirical probability measure* on a feature space. A sequence of token features $Z_N = \{z_i\}_{i=1}^N$ induces an empirical measure $\mu_N = \frac{1}{N}\sum_i \delta_{z_i}$ on a compact domain $\Omega \subset \mathbb{R}^d$. The KV cache is then a finite-support approximation $\tilde{\mu}$ with at most $m \ll N$ atoms. Under this view, KV compression is no longer a heuristic selection problem, but a *measure quantization* problem: construct a sparse measure that best approximates μ_N under a suitable metric.

This perspective is supported by recent theory. Transformers have been shown to be universal approximators of continuous functionals on spaces of probability measures (Furuya et al., 2025), and kernel mean embeddings provide an injective representation of distributions in reproducing kernel Hilbert spaces (Muandet et al., 2017; Sriperumbudur et al., 2010). At the same time, optimal transport (OT) and Wasserstein geometry offer natural metrics on probability measures (Villani, 2009; Cuturi, 2013), and classical quantization theory characterizes how well a measure on a d-dimensional manifold can be approximated by m atoms, with minimax error scaling as $m^{-1/d}$ (Graf & Luschgy, 2000). We combine these ideas to ask:

> *If Transformers are measure learners, what is the optimal way to compress the context measure under a fixed KV budget, and how does the induced ICL error scale with the budget and intrinsic dimension of the data?*

From Wasserstein Stability to Optimal KV Compression. Our key technical observation is that attention can be viewed as an integral operator over the context measure. Building on smoothness analyses of attention (Castin et al., 2024; Sander et al., 2022) and kernel embedding theory (Muandet et al., 2017; Sriperumbudur et al., 2010), we show that under mild regularity assumptions, Transformer ICL predictions are *Lipschitz* with respect to the Wasserstein-1 distance between context measures. This yields a geometric error bound: if a compressor $\mathcal{C}_m$ produces a sparse measure $\tilde{\mu}_m$ that is close to μ_N in Wasserstein-1, then the ICL prediction error is controlled by $\mathcal{W}_1(\mu_N, \tilde{\mu}_m)$. Combining this Lipschitz bound with classical quantization results (Graf & Luschgy, 2000; Villani, 2009) yields a minimax-optimal $m^{-1/d}$ scaling law for ICL error in the KV budget m, where d is the intrinsic feature dimension.

Contributions. We summarize our contributions as follows:

- **Theory: Transformers as Wasserstein-stable measure learners.** We formalize ICL as operator learning on the space of probability measures and, leveraging universality results (Furuya et al., 2025), show that Transformers with kernel attention approximate continuous functionals on $(\mathcal{P}(\Omega), \mathcal{W}_1)$. We prove that attention heads are Lipschitz with respect to the Wasserstein-1 distance, and derive a network-level error bound that directly links KV compression to ICL prediction error.
- **Optimal scaling laws for KV compression.** By coupling our Wasserstein stability bound with quantization theory (Graf & Luschgy, 2000), we show that if token features lie on a d-dimensional manifold, there exists a KV compression strategy such that the ICL error decays as $\tilde{\mathcal{O}}(m^{-1/d})$ in the KV budget m, and that this rate is minimax-optimal. This reveals an intrinsic-dimension bottleneck: structured scientific data (low d) is highly compressible, whereas high-entropy text is fundamentally harder to summarize.
- **Measure-Aware Compression (MAC).** Guided by this geometry, we introduce *Measure-Aware Compression* (MAC), a family of practical algorithms that explicitly respect the measure structure of the context. MAC-I performs post-hoc KV compression at inference time via weighted K-Means, approximating Wasserstein minimization, while MAC-T integrates Sinkhorn-regularized induced attention (Cuturi, 2013; Lee et al., 2019) into training to learn compressive representations end-to-end.
- **Empirical validation on synthetic and medical benchmarks.** On synthetic linear regression ICL tasks, MAC follows the predicted $m^{-1/d}$ scaling, whereas score-based pruning saturates early. On the LUNA16 pulmonary nodule detection benchmark (Setio et al., 2017), MAC achieves strong memory–accuracy tradeoffs, outperforming state-of-the-art KV compression baselines such as H2O and Quest (Zhang et al., 2024b; Tang et al., 2024) under the same memory budget. Qualitative analysis shows that MAC preserves rare but clinically critical "outlier" slices in feature space that heuristic methods tend to discard.

Overall, our results suggest that Transformers are not only sequence models, but *measure learners*, and that KV compression is best understood as constructing an optimal geometric summary of a context distribution rather than simply dropping low-score tokens.

2. Related Work

We organize related work into four areas: (i) theoretical views of In-Context Learning (ICL), (ii) KV cache compression and efficient long-context Transformers, (iii) optimal transport and kernel mean embeddings, and (iv) long-context applications in medical AI.

ICL Theory and Transformers as Operators. A growing line of work studies what and how Transformers can learn in context. Garg et al. (2022) analyze which simple function classes can be implemented via ICL, while Akyürek et al. (2023); Von Oswald et al. (2023); Dai et al. (2023) show that ICL can realize variants of gradient descent or meta-optimization, and Ahn et al. (2024); Mahankali et al. (2024) characterize linear self-attention as an optimal in-context learner under suitable assumptions. Mechanistic studies such as Olsson et al. (2022) further reveal "induction heads" that implement pattern completion in context. Most relevant to our perspective, Furuya et al. (2025) prove that Transformers are universal approximators for functionals on probability measures, suggesting that they can act directly on distributions rather than purely on ordered sequences. However, these works assume access to the full (uncompressed) context and do not address memory constraints. In contrast, we build on the measure-theoretic view of Furuya et al. (2025) and combine it with quantization theory to obtain Wasserstein-based stability results and minimax error scaling laws under KV compression.

KV Cache Compression and Efficient Long-Context Transformers. The quadratic cost of self-attention (Vaswani et al., 2017) and the linear growth of the KV cache motivate a large literature on efficient Transformers. Linear or streaming attention methods such as Katharopoulos et al. (2020) and state-space models like Mamba (Gu & Dao, 2024) reduce asymptotic complexity, yet empirical evidence suggests that standard Transformers remain highly competitive for ICL and long-context reasoning (Bietti et al., 2024). At the system level, architectures like LongRoPE, Unlimiformer, Gemini 1.5, and Llama 2 (Ding et al., 2024; Bertsch et al., 2024; Team et al., 2024; Touvron et al., 2023) push context lengths to millions of tokens, exacerbating the KV memory bottleneck.

KV cache compression methods can be roughly grouped into three categories. (1) *Token pruning*: H2O, Scissorhands, SnapKV, PyramidKV, and hybrid eviction schemes (Zhang et al., 2024b; Liu et al., 2024a; Li et al., 2024; Zhang et al., 2024a; Ge et al., 2024) retain "important" tokens according to accumulated attention scores or learned saliency, evicting the rest. (2) *Retrieval-based memory*: Quest, Infini-attention, RetrievalAttention, and related approaches (Tang et al., 2024; Munkhdalai et al., 2024; Gao et al., 2024; Bertsch et al., 2024) store a larger external memory and retrieve relevant blocks on demand, effectively trading KV size for retrieval latency and system complexity. (3) *Quantization and distillation*: KIVI, attention sinks, and distilled KV caches (Liu et al., 2024b; Xiao et al., 2024; Nawrot et al., 2024) compress each KV pair to a few bits or a smaller distilled representation without reducing the number of tokens. All of these methods operate primarily at the level of individual tokens or blocks and are guided by local importance scores. They do not model the context as a probability measure, and they provide no distribution-level guarantees that relate compression to ICL prediction error. Our work instead treats the KV cache as a finite-support approximation of an underlying context measure and casts KV compression as a measure quantization problem, which leads directly to geometric error bounds and intrinsic scaling laws.

Optimal Transport, Kernel Mean Embeddings, and Attention Geometry. Our analysis draws on tools from optimal transport (OT), kernel methods, and recent smoothness studies of attention. The Wasserstein distance and its computational approximations (Villani, 2009; Cuturi, 2013) provide natural metrics on probability measures and have been widely applied in modern machine learning. Classical quantization theory (Graf & Luschgy, 2000) characterizes how well a measure supported on a d-dimensional manifold can be approximated by a finite-support measure with m atoms, showing minimax rates of order $m^{-1/d}$. We leverage these results to derive intrinsic-dimension–dependent scaling laws for KV compression. Kernel mean embeddings (Muandet et al., 2017; Sriperumbudur et al., 2010) map probability measures into reproducing kernel Hilbert spaces and yield injective representations under characteristic kernels; this aligns naturally with viewing attention as an integral operator, and connects to Performer-style kernel approximations via random features (Song et al., 2021; Rahimi & Recht, 2007). The smoothness and Lipschitz properties of attention maps have been studied by Castin et al. (2024) and in doubly-stochastic attention variants such as Sinkformers (Sander et al., 2022), while Chizat & Bach (2018) analyze the dynamics of over-parameterized residual networks through an OT lens. Our work combines these ideas to show that attention is Lipschitz with respect to Wasserstein-1 and that KV compression error propagates through the network in a controlled way, yielding explicit Wasserstein-based error bounds.

Long-Context Modeling in Medical AI. Long-context reasoning is particularly important in medicine, where patient trajectories, multi-slice imaging studies, and multimodal records naturally form long sequences. Moor et al. (2023) highlight the need for generalist medical foundation models that can integrate such rich histories under realistic computational budgets. LUNA16 (Setio et al., 2017) provides a benchmark for pulmonary nodule detection from CT scans, where each study comprises hundreds of slices; this offers a natural testbed for evaluating long-context models and KV compression strategies in a clinically meaningful setting. By applying our measure-aware KV compression to LUNA16 and comparing against state-of-the-art baselines such as H2O and Quest (Zhang et al., 2024b; Tang et al., 2024), we demonstrate that geometric, distribution-level compression can preserve rare but clinically critical slices that heuristic pruning or retrieval tends to discard.

Summary. In summary, prior work has developed powerful operator-theoretic views of ICL and a rich toolkit of engineering solutions for KV compression, but lacks a unifying measure-based theory that explains what *optimal* KV compression should look like and how its error scales with the memory budget and intrinsic data dimension. Our contribution is to interpret Transformers as *measure learners*, to formalize KV compression as measure quantization under Wasserstein geometry, and to instantiate this view in practical algorithms that achieve state-of-the-art memory–accuracy tradeoffs on synthetic and medical long-context benchmarks.

3. Preliminaries

We briefly summarize the measure-theoretic and optimal transport notation used in our analysis, and formalize attention as an integral operator on probability measures.

Context as an Empirical Measure. We assume token features lie in a compact domain $\Omega \subset \mathbb{R}^d$. A length-$N$ context $Z_N = \{z_i\}_{i=1}^N$ (e.g., keys, values, or hidden states) induces an empirical probability measure

$$\mu_N = \frac{1}{N}\sum_{i=1}^{N}\delta_{z_i} \in \mathcal{P}(\Omega), \tag{1}$$

where δ_{z_i} is the Dirac mass at z_i and $\mathcal{P}(\Omega)$ denotes the set of Borel probability measures on Ω. Under this view, the KV cache is a finite-support approximation of μ_N with at most $m \ll N$ atoms; KV compression corresponds to replacing μ_N by a sparse measure $\tilde{\mu}_m$.

Wasserstein Distance and Quantization. For $p \geq 1$, the p-Wasserstein distance between $\mu, \nu \in \mathcal{P}(\Omega)$ is defined as (Villani, 2009)

$$W_p(\mu,\nu) := \left(\inf_{\pi\in\Pi(\mu,\nu)} \int_{\Omega\times\Omega} \|x-y\|^p \, d\pi(x,y)\right)^{1/p}, \tag{2}$$

where $\Pi(\mu,\nu)$ is the set of couplings with marginals μ and ν. We will focus on W_1, which metrizes weak convergence on $\mathcal{P}(\Omega)$ and admits a dual formulation that directly relates Lipschitz test functions to measure perturbations (Villani, 2009).

Given a budget m, a *quantization* (or compression) operator produces a discrete measure

$$\mathcal{C}_m(\mu_N) = \tilde{\mu}_m = \sum_{j=1}^{m} w_j\,\delta_{\tilde{z}_j}, \qquad \sum_{j=1}^{m} w_j = 1, \tag{3}$$

with at most m atoms. Classical quantization theory characterizes how small $W_1(\mu_N,\tilde{\mu}_m)$ can be as a function of m and the intrinsic dimension of the support of μ_N (Graf & Luschgy, 2000); we will leverage these results in our scaling-law analysis.

Kernel Mean Embeddings and Attention as Integration. Let $k : \Omega\times\Omega \to \mathbb{R}$ be a positive-definite kernel with associated reproducing kernel Hilbert space (RKHS) $\mathcal{H}_K$ (Muandet et al., 2017). The *kernel mean embedding* of a measure $\mu \in \mathcal{P}(\Omega)$ is

$$\phi(\mu) := \int_{\Omega} k(\cdot,z)\,d\mu(z) \in \mathcal{H}_K. \tag{4}$$

For characteristic kernels, this map is injective (Sriperumbudur et al., 2010), providing an implicit representation of μ as a point in $\mathcal{H}_K$.

We model a single attention head as an integral operator over the context measure. Given a query $q \in \mathbb{R}^{d_q}$, a kernel family $\kappa(q,\cdot)$, and a value map $\phi : \Omega \to \mathbb{R}^{d_v}$, we define

$$\mathrm{Attn}(q;\mu) := \frac{\int_\Omega \kappa(q,z)\,\phi(z)\,d\mu(z)}{\int_\Omega \kappa(q,z)\,d\mu(z)}. \tag{5}$$

Standard softmax attention (Vaswani et al., 2017) is recovered by choosing $\kappa(q,z) = \exp(q^\top k)$ and letting z bundle keys and values (k,v) with $\phi(z) = v$. In the discrete case $\mu = \mu_N$, the integrals in (5) reduce to finite sums over tokens; when μ is compressed to $\tilde{\mu}_m$, the same operator is applied to a sparse measure with m atoms.

This measure-based formulation makes the dependence of attention on μ explicit, and will allow us to study the Lipschitz continuity of $\mathrm{Attn}(\cdot;\mu)$ with respect to W_1 and to propagate KV compression error through multi-layer Transformers in Section 4.

4. Theoretical Framework

We now formalize Transformers as *measure learners* and derive the key ingredients of our framework: (i) universality on spaces of probability measures, (ii) Wasserstein stability of attention, and (iii) minimax scaling laws for KV compression.

Throughout, $\Omega \subset \mathbb{R}^d$ is compact, $\mathcal{P}(\Omega)$ is the set of Borel probability measures on Ω, and W_1 denotes the 1-Wasserstein distance on $\mathcal{P}(\Omega)$ (Villani, 2009). A context $Z_N = \{z_i\}_{i=1}^N$ induces an empirical measure μ_N, and a compressed KV cache corresponds to a sparse measure $\tilde{\mu}_m$ with at most m atoms.

4.1. Setup: ICL as Operator Learning on Measures

We model In-Context Learning as *operator learning* on measures. Given a context measure $\mu \in \mathcal{P}(\Omega)$ and a query $q \in \Omega$, the ideal in-context predictor is a functional

$$F : \mathcal{P}(\Omega) \times \Omega \to \mathcal{Y}, \qquad (\mu, q) \mapsto F(\mu, q), \tag{6}$$

where $\mathcal{Y}$ is the output space (e.g., logits, regression targets). Transformers with kernel attention (Eq. (5)) define a parametric family of such functionals, which we denote by $\Lambda_{\theta\theta}(\mu, q)$.

We make a mild regularity assumption on the feature space and kernel family.

Assumption 4.1 (Regularity)**.** 1. $\Omega \subset \mathbb{R}^d$ is compact.
2. The kernel family $\{\kappa(q, \cdot) : q \in \Omega\}$ is characteristic and Lipschitz, and the value map ϕ is bounded and Lipschitz.
3. For all relevant $\mu \in \mathcal{P}(\Omega)$ and $q \in \Omega$, the denominator in Eq. (5) is bounded below: $\int \kappa(q, z)\, d\mu(z) \geq \delta > 0$.

Assumption 4.1 is standard in kernel mean embedding and OT analyses (Muandet et al., 2017; Sriperumbudur et al., 2010; Villani, 2009; Castin et al., 2024) and holds for common choices such as Gaussian kernels on bounded domains.

4.2. Universality on Spaces of Measures

Recent work shows that Transformers are universal approximators for functionals on probability measures (Furuya et al., 2025). We adapt this result to our kernel-attention formulation.

Theorem 4.2 (Universal Approximation on Measures)**.** *Let $C(\mathcal{P}(\Omega) \times \Omega)$ denote the space of continuous functionals on $\mathcal{P}(\Omega) \times \Omega$, where $\mathcal{P}(\Omega)$ is equipped with the weak topology metrized by W_1. Under Assumption 4.1, for any $F \in C(\mathcal{P}(\Omega) \times \Omega)$ and any $\varepsilon > 0$, there exists a Transformer $\Lambda_{\theta\theta}$ with finitely many kernel-attention heads and MLP layers such that*

$$\sup_{\mu \in \mathcal{P}(\Omega),\, q \in \Omega} \left\| F(\mu, q) - \Lambda_{\theta\theta}(\mu, q) \right\| < \varepsilon. \tag{7}$$

The proof (Appendix A.2) combines the universality of Transformers on measures (Furuya et al., 2025) with the injectivity of kernel mean embeddings under characteristic kernels (Muandet et al., 2017; Sriperumbudur et al., 2010). Intuitively, attention heads generate a rich algebra of measure-dependent features that separates points in $\mathcal{P}(\Omega) \times \Omega$, and MLPs provide the final nonlinear mixing.

4.3. Wasserstein Stability of Attention

To understand KV compression, we need to quantify how sensitive attention is to perturbations of the context measure. We view a single attention head as the map $\mu \mapsto \mathrm{Attn}(q; \mu)$ defined in Eq. (5).

Lemma 4.3 (Attention is Lipschitz in W_1)**.** *Under Assumption 4.1, for any fixed query $q \in \Omega$, the attention operator $\mu \mapsto Attn(q; \mu)$ is Lipschitz with respect to W_1:*

$$\left\| Attn(q; \mu) - Attn(q; \nu) \right\| \;\leq\; L_{Attn}\, W_1(\mu, \nu), \quad \forall \mu, \nu \in \mathcal{P}(\Omega), \tag{8}$$

where L_{Attn} depends on the Lipschitz constants and bounds of κ and ϕ, the diameter of Ω, and the lower bound δ on the denominator.

The proof (Appendix A.3) uses the Kantorovich–Rubinstein duality for W_1 and a quotient-rule argument: both the numerator and denominator in Eq. (5) are Lipschitz linear functionals of μ, and the denominator is bounded away from zero, so their ratio is Lipschitz. This is consistent with recent smoothness analyses of attention maps (Castin et al., 2024; Sander et al., 2022).

4.4. Network-Level KV Compression Bound

KV compression replaces the full context measure μ_N by a sparse approximation $\tilde{\mu}_m = \mathcal{C}_m(\mu_N)$ with at most m atoms. We now show that, under mild Lipschitz assumptions on the layers, the induced ICL error is controlled by $W_1(\mu_N, \tilde{\mu}_m)$.

Let $\Lambda_{\theta\theta} = \Lambda^{(L)} \circ \cdots \circ \Lambda^{(1)}$ denote an L-layer Transformer. Assume each layer $\Lambda^{(l)}$ is $L^{(l)}_{\text{layer}}$-Lipschitz in its measure argument (this holds for attention + MLP + residual blocks under Assumption 4.1 and standard normalization).

Theorem 4.4 (KV Compression Error Bound)**.** *Under Assumption 4.1, let μ_N be the empirical context measure and $\tilde{\mu}_m = \mathcal{C}_m(\mu_N)$ any compressed measure with at most m atoms. Then*

$$\left\| \Lambda_{\theta\theta}(\mu_N, q) - \Lambda_{\theta\theta}(\tilde{\mu}_m, q) \right\| \;\leq\; L_{net}\, W_1(\mu_N, \tilde{\mu}_m), \tag{9}$$

where

$$L_{net} \;\coloneqq\; \prod_{l=1}^{L} L^{(l)}_{layer}. \tag{10}$$

The proof (Appendix A.4) simply composes the Lipschitz bound in Lemma 4.3 with the Lipschitz continuity of MLP and normalization layers. The constant L_{net} can in principle grow with depth, but techniques such as spectral normalization or Lipschitz regularization (e.g., Chizat & Bach, 2018; Castin et al., 2024) can be used to keep it controlled in practice. Crucially, Theorem 4.4 decouples the *geometric* difficulty of compression (how small can $W_1(\mu_N, \tilde{\mu}_m)$ be?) from the *architectural* sensitivity of the network (how large is L_{net}?).

4.5. Optimal Scaling Laws via Quantization

The remaining question is: how small can $W_1(\mu_N, \tilde{\mu}_m)$ be for a given KV budget m? This is exactly the object of classical quantization theory (Graf & Luschgy, 2000; Villani, 2009).

Corollary 4.5 (Minimax-Optimal Scaling for KV Compression)**.** *Suppose the support of μ_N lies on a d-dimensional submanifold of $\mathbb{R}^D$ with mild regularity (Graf & Luschgy, 2000). Then there exists a compression operator $\mathcal{C}_m$ that produces a measure $\tilde{\mu}_m$ with at most m atoms such that*

$$W_1(\mu_N, \tilde{\mu}_m) \;\le\; C\, m^{-1/d}, \tag{11}$$

for some constant C depending on the data distribution but not on m. Moreover, for general measures supported on d-dimensional manifolds, this rate $m^{-1/d}$ is minimax-optimal and cannot be improved in general. Combined with Theorem 4.4, the ICL prediction error under KV compression decays as

$$\big\|\Lambda_{\theta\theta}(\mu_N, q) - \Lambda_{\theta\theta}(\tilde{\mu}_m, q)\big\| \;=\; \tilde{\mathcal{O}}(m^{-1/d}), \tag{12}$$

where the $\tilde{\mathcal{O}}$ hides logarithmic factors and network-dependent Lipschitz constants.

Corollary 4.5 formalizes the *intrinsic-dimension bottleneck* for KV compression: contexts whose token features lie on low-dimensional manifolds (e.g., structured scientific or medical data) are highly compressible, whereas high-entropy text with large intrinsic dimension is fundamentally harder to summarize, regardless of algorithmic ingenuity.

This theory directly motivates our Measure-Aware Compression (MAC) algorithms in Section 5, which approximate the optimal quantizers predicted by Graf & Luschgy (2000) using weighted K-Means and Sinkhorn-regularized OT (Cuturi, 2013), and empirically achieve the predicted $m^{-1/d}$ scaling on synthetic tasks while improving memory–accuracy tradeoffs on real-world benchmarks.

5. Measure-Aware Compression (MAC)

The theory in Section 4 suggests that an optimal KV compressor should *(i)* minimize the Wasserstein-1 distance $W_1(\mu_N, \tilde{\mu}_m)$ between the full and compressed context measures, and *(ii)* exploit the intrinsic low-dimensional structure of token features to achieve the minimax $m^{-1/d}$ scaling. In practice, however, solving the exact W_1-optimal quantization problem is computationally prohibitive for large N.

We therefore introduce *Measure-Aware Compression* (MAC), a family of algorithms that approximate the Wasserstein minimizer under realistic compute budgets:

- **MAC-I** (Inference-only): a post-hoc KV compressor based on weighted K-Means, applied once during the prefill phase of generation.
- **MAC-T** (Training-time): an end-to-end learnable compressor that combines induced set attention (Lee et al., 2019) with a Sinkhorn-regularized OT loss (Cuturi, 2013).

Both variants treat the context as a probability measure and explicitly construct a finite-support approximation $\tilde{\mu}_m$ rather than scoring and dropping individual tokens.

5.1. MAC-I: Inference-Time Compression

For pre-trained models where weights are frozen, we compress the KV cache after prefill, before decoding. Let $K \in \mathbb{R}^{N\times D}$ and $V \in \mathbb{R}^{N\times D}$ denote the keys and values for a given layer and head (for simplicity we describe the single-head case).

Algorithm. MAC-I interprets the rows of K as samples from the context measure μ_N and seeks a finite-support approximation with m atoms. Since K-Means minimizes a discrete W_2^2-type quantization error, which upper-bounds W_1^2 on bounded domains (Villani, 2009; Graf & Luschgy, 2000), it provides a natural, scalable proxy for Wasserstein minimization.

Algorithm 1 MAC-I: Inference-Time KV Compression

1: **Input:** Keys $K \in \mathbb{R}^{N\times D}$, values $V \in \mathbb{R}^{N\times D}$, budget m.
2: **(Optional) Dimensionality reduction:** Project K to $K' \in \mathbb{R}^{N\times d_{\text{eff}}}$ via PCA or a learned linear map, with $d_{\text{eff}} \ll D$.
3: **K-Means clustering:** Run m-means on K' with K-Means++ seeding (Arthur & Vassilvitskii, 2007) for t iterations to obtain centroids $\{\tilde{k}_j\}_{j=1}^m$ and assignments $c(i) \in \{1,\dots,m\}$.
4: **Value aggregation:** For each cluster $C_j = \{i : c(i) = j\}$, define

$$\tilde{v}_j \leftarrow \frac{1}{|C_j|}\sum_{i\in C_j} V_i,$$
$$w_j \leftarrow \frac{|C_j|}{N}.$$

5: **Output:** Compressed KV cache $\{(\tilde{k}_j, \tilde{v}_j, w_j)\}_{j=1}^m$, representing the sparse measure $\tilde{\mu}_m = \sum_{j=1}^m w_j \delta_{(\tilde{k}_j, \tilde{v}_j)}$.

At decoding time, attention is computed against the compressed keys and values, respecting the cluster weights w_j (i.e., multiplying attention logits by w_j or equivalently repeating each centroid according to its mass). For multi-head attention, we either (a) run MAC-I independently per head (more accurate, higher cost) or (b) cluster in a shared low-dimensional space of concatenated keys to amortize cost across heads; we compare these options in Appendix B.

Complexity. For a single layer and head, MAC-I incurs $O(tNmd_{\text{eff}})$ time and $O(mD)$ memory during prefill. Since prefill is amortized over many decoding steps in long generation, this overhead is competitive with token-pruning methods that recompute scores at each step (Zhang et al., 2024b; Liu et al., 2024a; Li et al., 2024) and with retrieval-based schemes that repeatedly invoke a vector index (Tang et al., 2024; Gao et al., 2024). Once compressed, decoding operates on a KV cache of size m with the same $O(m)$ per-step complexity as other KV compression methods.

5.2. MAC-T: Training-Time Compression

When fine-tuning or domain-specific training is allowed (e.g., in medical or scientific settings), we can learn a compressive representation jointly with the task. MAC-T adds a small *measure-aware bottleneck* in front of standard attention.

Induced Set Attention as Learnable Quantizer. We adopt induced set attention blocks from the Set Transformer (Lee et al., 2019): a set of m learnable inducing points $I \in \mathbb{R}^{m\times D}$ attends to the full context Z_N and produces a compressed set $\{\tilde{z}_j\}_{j=1}^m$. This defines a parametric map $\mu_N \mapsto \tilde{\mu}_m^\theta = \frac{1}{m}\sum_{j=1}^m \delta_{\tilde{z}_j}$ that plays the role of $\mathcal{C}_m(\cdot)$ in our theory. We then feed $\tilde{\mu}_m^\theta$ to downstream attention layers instead of μ_N.

Sinkhorn-regularized OT Coverage. Without additional structure, inducing points may collapse onto a few modes and ignore rare but important regions of the measure. To encourage coverage of the support of μ_N, we add an entropic OT regularizer between the empirical context measure and the induced compressed measure:

$$\mathcal{L}_{\mathrm{OT}} = \mathrm{Sinkhorn}_\varepsilon(\mu_N, \tilde{\mu}_m^\theta), \tag{13}$$

where $\mathrm{Sinkhorn}_\varepsilon$ is the Sinkhorn approximation to W_1 with entropy parameter ε computed via iterative matrix scaling (Cuturi, 2013). The total training objective becomes

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \lambda\,\mathcal{L}_{\mathrm{OT}}, \tag{14}$$

with λ controlling the strength of the geometric regularization.

This design closely matches the structure predicted by our theory: the induced points play the role of quantization centers, and the Sinkhorn loss encourages the learned compressor to minimize a differentiable approximation of $W_1(\mu_N, \tilde{\mu}_m)$. In contrast to retrieval-based methods with non-differentiable selection (Tang et al., 2024; Gao et al., 2024), MAC-T is fully end-to-end trainable.

5.3. Practical Design Choices and Comparison

We summarize the computational profile of MAC against representative baselines below, where N is context length, m is KV budget, and D is hidden dimension:

Table 1. Asymptotic complexity per layer during prefill and decoding.

Method	Prefill	Decode (per step)
Full Attention	$O(N^2D)$	$O(ND)$
Token Pruning (H2O)	$O(N^2D)$	$O(mD)$
Retrieval (Quest)	$O(ND + \log N)$	$O(mD + \log N)$
Quantized KV (KIVI)	$O(N^2D)$	$O(ND)$
MAC-I (Ours)	$O(tNmD)$	$O(mD)$
MAC-T (Ours)	$O(N^2D)$	$O(mD)$

In MAC-I, the only additional cost over standard prefill is the $O(tNmD)$ clustering step, which is typically dominated by the $O(N^2)$ attention cost when $m \ll N$ and t is small (we use $t \le 10$ in our experiments). In MAC-T, we pay the same asymptotic cost as standard attention during training but replace the full context by m inducing tokens at inference time.

Conceptually, MAC differs from existing KV compression strategies in how it uses the geometry of the context: rather than scoring and dropping tokens, it *constructs* a small set of representative atoms that approximate the full context measure in Wasserstein geometry. Section 6 shows that this measure-aware design is sufficient to recover the predicted $m^{-1/d}$ scaling on synthetic tasks and to outperform strong baselines such as H2O and Quest on the LUNA16 medical benchmark.

6. Experiments

We empirically evaluate MAC on (i) a controlled synthetic regression task, where we can directly test the $m^{-1/d}$ scaling predicted by Corollary 4.5, and (ii) a real-world long-context medical benchmark (LUNA16) that reflects the intended scientific use cases of our framework (Moor et al., 2023; Setio et al., 2017). Unless otherwise noted, we apply MAC-I at inference time; additional details, including MAC-T results, are deferred to Appendix B.

6.1. Experimental Setup

Models. All experiments use standard Transformer encoders with multi-head attention (Vaswani et al., 2017). For each setting, we first train a baseline model on full contexts (no KV compression), then freeze the weights and apply different KV compression schemes only at inference time. This isolates the effect of compression from model capacity. We report mean performance over multiple random seeds.

Baselines. We compare MAC against:

- **Full KV**: no compression (upper bound).
- **Random**: keep m tokens chosen uniformly at random.
- **H2O** (Zhang et al., 2024b): heavy-hitter token pruning

Figure 1. **Synthetic linear regression.** Test MSE versus KV budget m (log–log scale). MAC-I tracks the predicted $m^{-1/d}$ slope, while score-based pruning (H2O) saturates for moderate m.

based on accumulated attention scores.
- **Quest** (Tang et al., 2024): query-aware page-level sparsification.

For fairness, all compressed methods use the same KV budget m at test time.

6.2. Synthetic Linear Regression: Testing the Scaling Law

Task. We follow the in-context linear regression setup used in prior ICL work (e.g. Garg et al., 2022; Akyürek et al., 2023; Von Oswald et al., 2023). Inputs $x_i \in \mathbb{R}^d$ are drawn from a mixture of Gaussians, outputs are $y_i = w^\top x_i + \epsilon$, and the model must predict y for new queries given a context of N labeled examples. Here d is controlled, allowing us to probe the intrinsic-dimension dependence in Corollary 4.5.

Protocol. We train a 4-layer Transformer encoder (no positional encodings) on full contexts of length $N = 1024$. At test time, we compress the KV cache to size $m \in \{8, 16, 32, 64\}$ using MAC-I, Random, and H2O, and measure the mean-squared error (MSE) on held-out queries. MAC-I uses a low-dimensional projection of keys ($d_{\text{eff}} \ll D$) before K-Means, as described in Section 5.1.

Results. Figure 1 shows MSE versus m on a log–log scale. MAC-I closely follows a straight line with slope consistent with the theoretical $\mathcal{O}(m^{-1/d})$ prediction, while Random pruning decays much more slowly and H2O plateaus early. Intuitively, H2O over-focuses on tokens that are salient for the current query but fails to preserve a good geometric summary of the underlying data distribution, whereas MAC explicitly approximates the context measure.

6.3. LUNA16 Pulmonary Nodule Detection

Task and Data. LUNA16 is a public benchmark for pulmonary nodule detection from chest CT scans (Setio et al., 2017). Each scan consists of 200–500 slices; following Moor et al. (2023), we view a study as a long sequence of 2D slices and ask the model to predict, at study level, whether it contains nodules. This is a prototypical long-context medical task where only a few slices (nodules) are clinically critical, and the rest provide background.

Model and Features. We extract per-slice visual features using a ResNet-50 backbone pretrained on ImageNet, projected to a D-dimensional embedding. A 4-layer Transformer encoder with multi-head attention processes the sequence, and a pooled representation feeds a binary classifier. The model is trained on full sequences; at test time we apply MAC-I, H2O, Quest, or windowed attention under the same KV budget m.

Results. Table 2 reports accuracy, peak VRAM usage, and relative decoding speed for the LUNA16 test set with a fixed budget of $m = 64$ cached tokens (roughly 10–20% of the original context length).

Table 2. LUNA16 nodule detection with KV budget $m = 64$.

METHOD	ACCURACY	VRAM	SPEEDUP
FULL KV	94.5%	24GB	1.0×
LOCAL WINDOW	78.2%	**2GB**	4.5×
H2O (ZHANG ET AL., 2024B)	85.1%	4GB	4.2×
QUEST (TANG ET AL., 2024)	88.4%	3GB	4.3×
MAC-I (OURS)	**92.3%**	4GB	4.1×

MAC-I achieves substantially better accuracy than H2O and Quest at similar memory and speed, closing most of the gap to the Full KV upper bound while reducing VRAM by $\sim 6\times$. Qualitative inspection of the learned clusters shows that MAC tends to assign dedicated centroids to rare nodule-like slices in feature space, whereas H2O sometimes evicts them in favor of more frequently attended but less informative slices.

6.4. Summary

Across synthetic and real-world settings, the empirical behavior of MAC closely matches our measure-theoretic analysis: (i) on controlled regression tasks, MAC follows the $m^{-1/d}$ error decay predicted by quantization theory; (ii) on LUNA16, MAC delivers state-of-the-art memory–accuracy tradeoffs compared to strong KV compression baselines, supporting the view that KV compression is best understood as constructing an optimal geometric summary of the context distribution rather than dropping low-score tokens.

7. Limitations and Discussion

Our results support the view that Transformers behave as *measure learners* and that KV compression is fundamentally a quantization problem in Wasserstein geometry. At the same time, the framework has several limitations that point to concrete directions for future work.

Computational Cost of Measure-Aware Compression. MAC-I trades off better geometric fidelity for additional prefill cost: K-Means-based clustering scales as $O(tNmD)$, which is cheaper than full $O(N^2D)$ attention but more expensive than simple top-k token scoring (Zhang et al., 2024b; Liu et al., 2024a) or scalar quantization (Liu et al., 2024b). In the settings we study, this overhead is amortized over long decoding horizons, but for very short sequences or extremely tight latency constraints, lighter-weight approximations (e.g., hierarchical clustering or coresets (Ge et al., 2024), or retrieval-style vector indices (Gao et al., 2024)) may be preferable. A promising direction is to co-design MAC with the underlying inference stack so that geometric compression reuses infrastructure already deployed for retrieval (Tang et al., 2024; Bertsch et al., 2024).

From Static Measures to Causal, Infinite Context. Our theory treats the context as a static measure and is explicitly permutation invariant. In contrast, autoregressive LMs and streaming models operate with a time-evolving context μ_t and causal masking. Extending our Wasserstein stability results to this setting likely requires "time-lifting" or path-space constructions (Furuya et al., 2025; Castin et al., 2024), and a careful treatment of recurrent or infinite-context architectures such as Infini-attention (Munkhdalai et al., 2024; Bertsch et al., 2024) and state-space models (Gu & Dao, 2024). Developing measure-theoretic compression schemes that remain consistent along the entire generation trajectory is an important open problem.

Tightness of Lipschitz Bounds. The KV compression error bound in Theorem 4.4 is worst-case and depends on a global network Lipschitz constant L_{net}, which can be loose for deep Transformers. In practice, normalization, residual connections, and attention structure often induce much smaller effective sensitivities than the naive product of per-layer constants would suggest (Castin et al., 2024; Sander et al., 2022; Chizat & Bach, 2018). Characterizing the *effective* Lipschitz behavior of trained models and tying it to their KV compressibility is an interesting direction, and could motivate explicit regularization schemes that improve both robustness and compressibility.

Scope of Empirical Validation. Our experiments focus on medium-scale Transformers and two representative settings: synthetic linear regression and long-context medical imaging (Moor et al., 2023; Setio et al., 2017). While this is sufficient to observe the predicted $m^{-1/d}$ scaling and to outperform strong KV compression baselines, it does not yet demonstrate benefits at the scale of frontier language models (Touvron et al., 2023; Team et al., 2024; Ding et al., 2024). Scaling MAC to multi-billion parameter models and diverse domains (e.g., code, multi-modal scientific data) is primarily an engineering challenge but may also reveal new phenomena not captured by our current theory.

Interaction with Other Efficiency Techniques. We study MAC on top of standard softmax attention (Vaswani et al., 2017). In practice, KV compression is often combined with linear/streaming attention (Katharopoulos et al., 2020; Xiao et al., 2024), quantization (Liu et al., 2024b), and retrieval (Tang et al., 2024; Gao et al., 2024). Understanding how measure-aware compression composes with these techniques—e.g., whether MAC can serve as a geometric backbone into which retrieval or quantization plug as special cases—is an exciting direction for building unified long-context systems.

8. Conclusion

We proposed a measure-theoretic view of In-Context Learning in Transformers, modeling the KV cache as a finite-support approximation of a context measure. Under mild regularity assumptions, we showed that kernel attention is Lipschitz with respect to the Wasserstein-1 distance, and combined this with classical quantization theory to derive a minimax-optimal $m^{-1/d}$ scaling law for KV compression error. Guided by this geometry, we introduced Measure-Aware Compression (MAC), which approximates Wasserstein-optimal quantization via K-Means at inference time and Sinkhorn-regularized induced attention during training. MAC empirically follows the predicted scaling on synthetic regression and achieves favorable memory–accuracy tradeoffs on a real-world medical benchmark compared to state-of-the-art KV compression methods.

More broadly, our results support the thesis that Transformers are not only sequence models but *measure learners*, and that KV compression is best understood as constructing an optimal geometric summary of the context distribution rather than dropping low-score tokens. We hope this perspective will help bridge theory and systems for long-context models, and inspire new compression and memory architectures grounded in optimal transport and the geometry of probability measures.

Broader Impact

Our work is methodological: we study Transformers as measure learners and design geometric KV compression schemes. A positive consequence is that long-context models may become more memory- and energy-efficient, making it easier for resource-limited settings (e.g., hospitals or smaller labs) to use foundation models on long medical or scientific sequences (Moor et al., 2023; Setio et al., 2017; Team et al., 2024; Ding et al., 2024; Gu & Dao, 2024).

At the same time, cheaper and more scalable long-context inference also lowers the barrier for misuse, for example in privacy-invasive logging or more targeted generation of harmful content. Our methods do not address dataset biases, fairness, or privacy by themselves; they should be combined with appropriate data governance and safety measures (Vaswani et al., 2017; Touvron et al., 2023). Overall, we view our contribution as clarifying the geometry of in-context learning; its societal impact will depend on how it is instantiated and regulated in downstream applications.

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A. Detailed Proofs of Theoretical Results

In this appendix we provide proofs for the main theoretical results in Section 4. Throughout, $\Omega \subset \mathbb{R}^d$ is a compact metric space, $\mathcal{P}(\Omega)$ is the set of Borel probability measures on Ω, and W_1 denotes the 1-Wasserstein distance on $\mathcal{P}(\Omega)$ (Villani, 2009).

A.1. Notation and Preliminaries

For a bounded measurable function $f : \Omega \to \mathbb{R}^k$ and $\mu \in \mathcal{P}(\Omega)$, we write

$$\langle f, \mu \rangle \;:=\; \int_\Omega f(z)\, d\mu(z).$$

The Kantorovich–Rubinstein dual formulation of W_1 states that for all $\mu, \nu \in \mathcal{P}(\Omega)$,

$$W_1(\mu, \nu) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left| \langle f, \mu \rangle - \langle f, \nu \rangle \right|, \tag{15}$$

where the supremum runs over all 1-Lipschitz real-valued functions on Ω (Villani, 2009).

Let $\mathcal{H}_K$ be the reproducing kernel Hilbert space (RKHS) associated with a positive definite kernel $\kappa : \Omega \times \Omega \to \mathbb{R}$. The kernel mean embedding of $\mu \in \mathcal{P}(\Omega)$ is

$$\phi(\mu) \;:=\; \int_\Omega \kappa(\cdot, z)\, d\mu(z) \in \mathcal{H}_K.$$

A kernel is *characteristic* if the map $\mu \mapsto \phi(\mu)$ is injective; see Muandet et al. (2017); Sriperumbudur et al. (2010).

A.2. Proof of Theorem 4.2 (Universality on Measures)

Restatement. Under Assumption 4.1, for any continuous functional $F \in C(\mathcal{P}(\Omega) \times \Omega)$ and any $\varepsilon > 0$, there exists a Transformer Λ_θ with kernel-attention heads and MLP layers such that

$$\sup_{\mu \in \mathcal{P}(\Omega), q \in \Omega} \left\| F(\mu, q) - \Lambda_\theta(\mu, q) \right\| < \varepsilon.$$

Proof sketch. We follow the strategy of Furuya et al. (2025), adapting it to our kernel-attention parameterization.

1. Compactness. By Assumption 4.1, Ω is compact. By Prokhorov's theorem, $\mathcal{P}(\Omega)$ is compact under the topology of weak convergence, which is metrized by W_1 on bounded domains (Villani, 2009). Therefore $\mathcal{X} := \mathcal{P}(\Omega) \times \Omega$ is compact Hausdorff.

2. Function algebra generated by attention heads. Consider the class $\mathcal{A}$ of functions from $\mathcal{X}$ to $\mathbb{R}^k$ representable by a finite composition of: (i) kernel-attention maps of the form

$$(\mu, q) \mapsto \mathrm{Attn}(q; \mu) = \frac{\int \kappa(q, z)\, \varphi(z)\, d\mu(z)}{\int \kappa(q, z)\, d\mu(z)}$$

with bounded Lipschitz φ; and (ii) finite-width MLPs with continuous activations applied to finite collections of such features and q. Because MLPs are universal on compact subsets of Euclidean space, $\mathcal{A}$ is closed under finite linear combinations and pointwise products, and contains constant functions. Hence $\mathcal{A}$ is a subalgebra of $C(\mathcal{X})$.

3. Separation of points. Let $(\mu, q) \neq (\nu, r)$ in $\mathcal{X}$. We must show there exists $f \in \mathcal{A}$ such that $f(\mu, q) \neq f(\nu, r)$.

Case 1: $q \neq r$. Use an MLP acting only on the query: the map $q \mapsto \mathrm{MLP}(q)$ is universal on compact Ω, so there exists an MLP that separates q and r.

Case 2: $q = r$ and $\mu \neq \nu$. Consider the kernel mean embeddings

$$\phi(\mu) = \int \kappa(\cdot, z)\, d\mu(z), \qquad \phi(\nu) = \int \kappa(\cdot, z)\, d\nu(z).$$

Since κ is characteristic, $\phi(\mu) \neq \phi(\nu)$ (Muandet et al., 2017; Sriperumbudur et al., 2010). Therefore there exists some evaluation point $q^\star \in \Omega$ such that

$$\int \kappa(q^\star, z)\, d\mu(z) \;\neq\; \int \kappa(q^\star, z)\, d\nu(z).$$

This evaluation is precisely the (unnormalized) numerator of an attention head with query $q^\star$ and value map $\varphi \equiv 1$. Thus an attention head yields a scalar feature that separates μ and ν; combining it with a simple linear readout gives a function in $\mathcal{A}$ that separates $(\mu, q^\star)$ and $(\nu, q^\star)$.

Therefore $\mathcal{A}$ separates points in $\mathcal{X}$.

4. Stone–Weierstrass. By the Stone–Weierstrass theorem for compact Hausdorff spaces, any subalgebra $\mathcal{A}$ of $C(\mathcal{X})$ that contains constants and separates points is dense in $C(\mathcal{X})$ under the uniform norm. Hence for any $F \in C(\mathcal{X})$ and $\varepsilon > 0$, there exists $f \in \mathcal{A}$ with $\sup_{(\mu,q)\in\mathcal{X}} \|F(\mu, q) - f(\mu, q)\| < \varepsilon$.

Any such f can be realized by a Transformer with finitely many kernel-attention heads and MLP layers, since $\mathcal{A}$ is defined exactly as the set of functions realized by this architecture. This concludes the proof. □

A.3. Proof of Lemma 4.3 (Lipschitz Continuity)

Restatement. Under Assumption 4.1, for any fixed query $q \in \Omega$, the map $\mu \mapsto \mathrm{Attn}(q; \mu)$ is Lipschitz with respect to W_1:

$$\big\|\mathrm{Attn}(q;\mu) - \mathrm{Attn}(q;\nu)\big\| \le L_{\mathrm{Attn}}\, W_1(\mu,\nu), \quad \forall \mu,\nu \in \mathcal{P}(\Omega).$$

Proof. Fix q and define the scalar- or vector-valued functions

$$f_N(z) := \kappa(q,z)\,\varphi(z), \qquad f_D(z) := \kappa(q,z),$$

so that

$$N(\mu) := \langle f_N, \mu\rangle, \qquad D(\mu) := \langle f_D, \mu\rangle, \qquad \mathrm{Attn}(q;\mu) = \frac{N(\mu)}{D(\mu)}.$$

By Assumption 4.1, κ and φ are bounded and Lipschitz on compact Ω, hence f_N and f_D are bounded and Lipschitz. Let $\|f_N\|_\infty \le B_N$, $\|f_D\|_\infty \le B_D$, and $\mathrm{Lip}(f_N) \le L_N$, $\mathrm{Lip}(f_D) \le L_D$.

Using the dual formulation of W_1,

$$\|N(\mu) - N(\nu)\| = \left\| \int f_N\, d(\mu - \nu) \right\| \le L_N\, W_1(\mu,\nu),$$

and similarly

$$|D(\mu) - D(\nu)| \le L_D\, W_1(\mu,\nu).$$

Moreover, $|D(\mu)| \le B_D$ for all μ and, by Assumption 4.1, there exists $\delta > 0$ such that $D(\mu) \ge \delta$ for all relevant μ.

We now bound the difference of the quotients:

$$\begin{aligned}
\mathrm{Attn}(q;\mu) - \mathrm{Attn}(q;\nu) &= \frac{N(\mu)}{D(\mu)} - \frac{N(\nu)}{D(\nu)} \\
&= \frac{N(\mu)D(\nu) - N(\nu)D(\mu)}{D(\mu)D(\nu)} \\
&= \frac{N(\mu)\big(D(\nu) - D(\mu)\big) + D(\mu)\big(N(\mu) - N(\nu)\big)}{D(\mu)D(\nu)}.
\end{aligned}$$

Taking norms and using $|D(\mu)|, |D(\nu)| \ge \delta$,

$$\begin{aligned}
\big\|\mathrm{Attn}(q;\mu) - \mathrm{Attn}(q;\nu)\big\| &\le \frac{\|N(\mu)\|\,|D(\nu) - D(\mu)| + |D(\mu)|\,\|N(\mu) - N(\nu)\|}{|D(\mu)D(\nu)|} \\
&\le \frac{B_N L_D + B_D L_N}{\delta^2}\, W_1(\mu,\nu).
\end{aligned}$$

Thus the map is Lipschitz with constant $L_{\mathrm{Attn}} := (B_N L_D + B_D L_N)/\delta^2$. □

A.4. Proof of Theorem 4.4 (KV Compression Bound)

Restatement. Let $\Lambda_\theta = \Lambda^{(L)} \circ \cdots \circ \Lambda^{(1)}$ be an L-layer Transformer, where each layer $\Lambda^{(l)}$ is $L^{(l)}_{\text{layer}}$-Lipschitz in its measure argument. For any empirical context measure μ_N and compressed measure $\tilde{\mu}_m$,

$$\left\|\Lambda_\theta(\mu_N, q) - \Lambda_\theta(\tilde{\mu}_m, q)\right\| \le L_{\text{net}}\, W_1(\mu_N, \tilde{\mu}_m),$$

with $L_{\text{net}} = \prod_{l=1}^{L} L^{(l)}_{\text{layer}}$.

Proof. Define $h^{(0)}(\mu, q) = (\mu, q)$ and for $l \ge 1$,

$$h^{(l)}(\mu, q) \;=\; \Lambda^{(l)}\big(h^{(l-1)}(\mu, q)\big).$$

Then $\Lambda_\theta(\mu, q) = h^{(L)}(\mu, q)$.

Let $\mu = \mu_N$ and $\nu = \tilde{\mu}_m$. By Lipschitzness of each layer,

$$\left\|h^{(1)}(\mu, q) - h^{(1)}(\nu, q)\right\| \le L^{(1)}_{\text{layer}}\, W_1(\mu, \nu).$$

Applying the same argument recursively,

$$\begin{aligned}\left\|h^{(2)}(\mu, q) - h^{(2)}(\nu, q)\right\| &\le L^{(2)}_{\text{layer}} \left\|h^{(1)}(\mu, q) - h^{(1)}(\nu, q)\right\| \\ &\le L^{(2)}_{\text{layer}} L^{(1)}_{\text{layer}}\, W_1(\mu, \nu),\end{aligned}$$

and, by induction,

$$\left\|h^{(L)}(\mu, q) - h^{(L)}(\nu, q)\right\| \le \Big(\prod_{l=1}^{L} L^{(l)}_{\text{layer}}\Big)\, W_1(\mu, \nu).$$

Setting $\mu = \mu_N$ and $\nu = \tilde{\mu}_m$ yields the stated bound. □

A.5. Proof of Corollary 4.5 (Scaling Law)

Restatement. If the support of μ_N lies on a d-dimensional submanifold of $\mathbb{R}^D$ with mild regularity, there exists a compression operator producing $\tilde{\mu}_m$ with at most m atoms such that

$$W_1(\mu_N, \tilde{\mu}_m) \le C\, m^{-1/d},$$

and this rate is minimax-optimal in general. Combining with Theorem 4.4 yields an ICL prediction error of order $\tilde{\mathcal{O}}(m^{-1/d})$.

Proof sketch. The claim is a direct application of classical quantization theory for probability measures (Graf & Luschgy, 2000; Villani, 2009). For completeness, we briefly outline the relevant result.

Let $\mathcal{Q}_m$ be the set of all probability measures with support size at most m. Define the optimal L^1 quantization error of μ at level m as

$$e_1(\mu, m) \;:=\; \inf_{\nu \in \mathcal{Q}_m} W_1(\mu, \nu).$$

Under mild assumptions on μ—in particular, that it admits a density with respect to the d-dimensional Hausdorff measure on a compact d-dimensional manifold—there exist constants $0 < c \le C < \infty$ such that

$$c\, m^{-1/d} \le e_1(\mu, m) \le C\, m^{-1/d} \quad \text{for all sufficiently large } m.$$

The upper bound is achieved by appropriately chosen m-point quantizers; the lower bound shows that no other scheme can improve the rate uniformly over a rich class of measures (Graf & Luschgy, 2000).

Applying this to $\mu = \mu_N$ yields the claimed rate $W_1(\mu_N, \tilde{\mu}_m) \le C m^{-1/d}$ for some (possibly data-dependent) C. Plugging into Theorem 4.4 gives

$$\left\|\Lambda_\theta(\mu_N, q) - \Lambda_\theta(\tilde{\mu}_m, q)\right\| \le L_{\text{net}}\, C\, m^{-1/d},$$

which we write as $\tilde{\mathcal{O}}(m^{-1/d})$ to emphasize that L_{net} and logarithmic factors are absorbed into the constant. □

B. Experimental Details and Additional Results

We provide additional details on the experimental setups in Section 6, as well as ablations and implementation notes.

B.1. Synthetic Linear Regression

Data generation. We consider input dimension $d \in \{10, 20\}$ to probe the dependence on intrinsic dimension. For each task instance, we sample a weight vector $w \sim \mathcal{U}(-1, 1)^d$ and generate a context of $N = 1024$ pairs (x_i, y_i):

- x_i is drawn from a mixture of three Gaussians on $\mathbb{R}^d$, with randomly sampled means and shared covariance $0.1^2 I$. This yields a non-uniform, low-dimensional data manifold.
- $y_i = w^\top x_i + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 0.05$.

We generate fresh task instances at test time to avoid overfitting to a fixed w.

Model architecture. We use a 4-layer Transformer encoder with hidden dimension $D = 128$, 4 attention heads, and GELU activations. No positional encodings are used so that the model is encouraged to treat the context as a set (Lee et al., 2019). We prepend a dedicated query token and train the model to predict its target value given the rest of the context.

Training and evaluation. The model is trained with Adam for 10 000 steps, batch size 64, learning rate 10^{-3}, using full KV cache. At evaluation time, we freeze weights and apply KV compression with different budgets m. For MAC-I, we project keys to $d_{\text{eff}} = 16$ dimensions using a learned linear map, then run Mini-Batch K-Means with K-Means++ initialization (Arthur & Vassilvitskii, 2007). We average results over 5 seeds.

B.2. LUNA16 Pulmonary Nodule Detection

Preprocessing. We follow the LUNA16 protocol (Setio et al., 2017). CT volumes are resampled to a fixed voxel spacing, clipped to a standardized Hounsfield range, and normalized. We then extract 2D axial slices and center-crop each slice to a fixed resolution. Each study typically yields $N \approx$ 200–500 slices.

Feature extractor and Transformer. We use a ResNet-50 backbone pretrained on ImageNet to extract per-slice feature vectors from the penultimate layer (2048-dim). A linear projection maps these to a Transformer hidden dimension $D = 256$. We stack a 4-layer Transformer encoder with 8 heads and apply max-pooling over the sequence followed by a 2-layer MLP for binary classification.

Training. We train on full sequences using cross-entropy loss, with AdamW optimizer, learning rate 10^{-4}, batch size 16, and cosine learning rate schedule. We use early stopping based on validation accuracy.

MAC-I and baselines. At test time:

- **MAC-I**: we apply PCA to reduce key dimension to $d_{\text{eff}} = 32$ and run Mini-Batch K-Means with $m = 64$ clusters and $t = 10$ iterations. Cluster centroids and masses are used as compressed KV states.
- **H2O**: we implement heavy-hitter token selection as in Zhang et al. (2024b), accumulating attention scores over layers and pruning to the top-m tokens.
- **Quest**: we divide the sequence into pages of 16 slices and apply query-aware page selection following Tang et al. (2024).
- **Local window**: we restrict attention to a symmetric local window of size m around the query.

We report accuracy, peak GPU memory, and per-step decoding time averaged over the test set. All methods use the same underlying model and differ only in KV handling.

B.3. MAC-T Implementation Details

Induced set attention. For MAC-T experiments (not shown in the main text due to space), we insert an induced set attention block (Lee et al., 2019) between the feature extractor and the first Transformer layer. The block uses $m = 64$ inducing tokens $I \in \mathbb{R}^{m \times D}$ and produces a compressed set $\{\tilde{z}_j\}_{j=1}^m$ that replaces the original sequence.

Sinkhorn loss. We approximate the OT distance between the empirical measure over features $\mu_N = \frac{1}{N}\sum_{i=1}^{N}\delta_{z_i}$ and the compressed measure $\tilde{\mu}_m^\theta = \frac{1}{m}\sum_{j=1}^{m}\delta_{\tilde{z}_j}$ using the entropic Sinkhorn divergence (Cuturi, 2013) with regularization $\varepsilon = 0.05$ and 10 iterations. The OT term is added to the task loss with weight $\lambda = 0.1$.

Observed behavior. MAC-T typically learns inducing tokens that cover different semantic regions of feature space (e.g., normal lung tissue, nodule-like patterns), and achieves slightly better accuracy than MAC-I at the same budget m, at the cost of additional training overhead. The qualitative behavior is consistent with our interpretation of the inducing tokens as learned quantization centers guided by OT geometry.

B.4. Additional Ablations

Due to space, we summarize key trends:

- **Effect of intrinsic dimension.** On synthetic regression, increasing the input dimension d steepens the slope of the error curve in log–log space for all methods, but MAC-I remains closest to the theoretical $m^{-1/d}$ prediction, while Random and H2O deviate more strongly as d grows.
- **Budget sensitivity.** On LUNA16, MAC-I maintains a smooth accuracy degradation as m decreases, degrading gracefully down to $m = 32$. H2O and Quest exhibit sharper drops once m falls below $\approx 10\%$ of the original context.
- **Head-wise vs. shared clustering.** For moderate budgets, clustering in a shared key space across heads achieves nearly the same accuracy as per-head clustering, while being cheaper. At very small budgets, per-head clustering yields a modest but consistent improvement, suggesting some head-specific geometry.

These results further support the view that explicitly respecting the measure structure of the context—instead of operating on tokens independently—is crucial for robust KV compression in both synthetic and real-world long-context settings.